Appendix A - Geographe Marine Park Natural Values Report

South-west Marine Parks Network

Table of contents

Natural Values **1**
- Sampling summary . . . 2
- Benthic ecosystem extent . . . 3
  - Observations . . . 3
  - Dominant habitat types . . . 4
  - Habitat distribution and probability . . . 5
  - Functional reef distribution and probability . . . 6
- Benchmarks of benthic natural values . . . 7
  - Seagrass . . . 7
  - Macroalgae . . . 8
  - Sessile invertebrates . . . 9
- Benchmarks of demersal fish natural values . . . 10
  - Survey effort . . . 10
  - Species richness . . . 11
  - Total abundance . . . 13
  - Large Reef Fish Index* . . . 15
  - Reef fish thermal index . . . 17
- Threatened species . . . 19
  - Threatened shark and ray species . . . 20
  - Other shark and ray species . . . 21

Natural Values

Understanding of natural values

Sampling summary

Table A.1: Summary of monitoring campaigns in the study area, showing the survey dates, sampling method, management zones sampled, and the number of samples with each data type. NPZ = National Park Zone, HPZ = Habitat Protection Zone, MUZ = Multiple Use Zone, SPZ = Special Purpose Zone, SZ = Sanctuary Zone, GUZ = General Use Zone.

					Habitat	Fish	
Date	**Campaign name**	**Method**	**Areas**	**Samples**	**Count**	**Count**	**Length**
Dec 2014	2014-12 Geographe-bay stereo-BRUVs	BRUV	Coastal waters, GUZ, HPZ, MUZ, NPZ, SPZ, SZ	324	320	297	238
Mar-Apr 2024	2024-04 Geographe stereo-BRUVs	BRUV	Coastal waters, GUZ, HPZ, MUZ, NPZ, SPZ, SZ	261	260	259	246
Apr 2024	2024-04 Geographe BOSS	BOSS	GUZ, HPZ, MUZ, NPZ	121	121	-	-

Benthic ecosystem extent

Observations

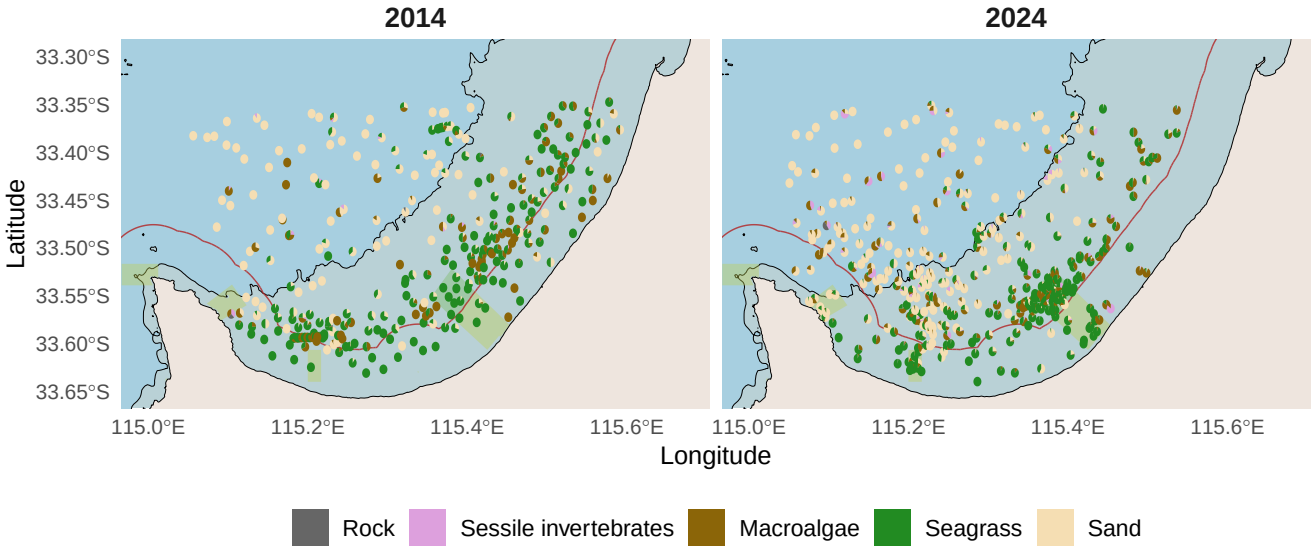


Figure A.1: Habitat classes observed in stereo-BRUV imagery. Habitat classes observed in spatially balanced stereo-BRUV and drop-camera imagery visualised as spatial pie charts. State and Commonwealth marine park boundaries are shown. The red line delimits state and commonwealth waters. Bathymetric contour at 30 m is shown.

Dominant habitat types

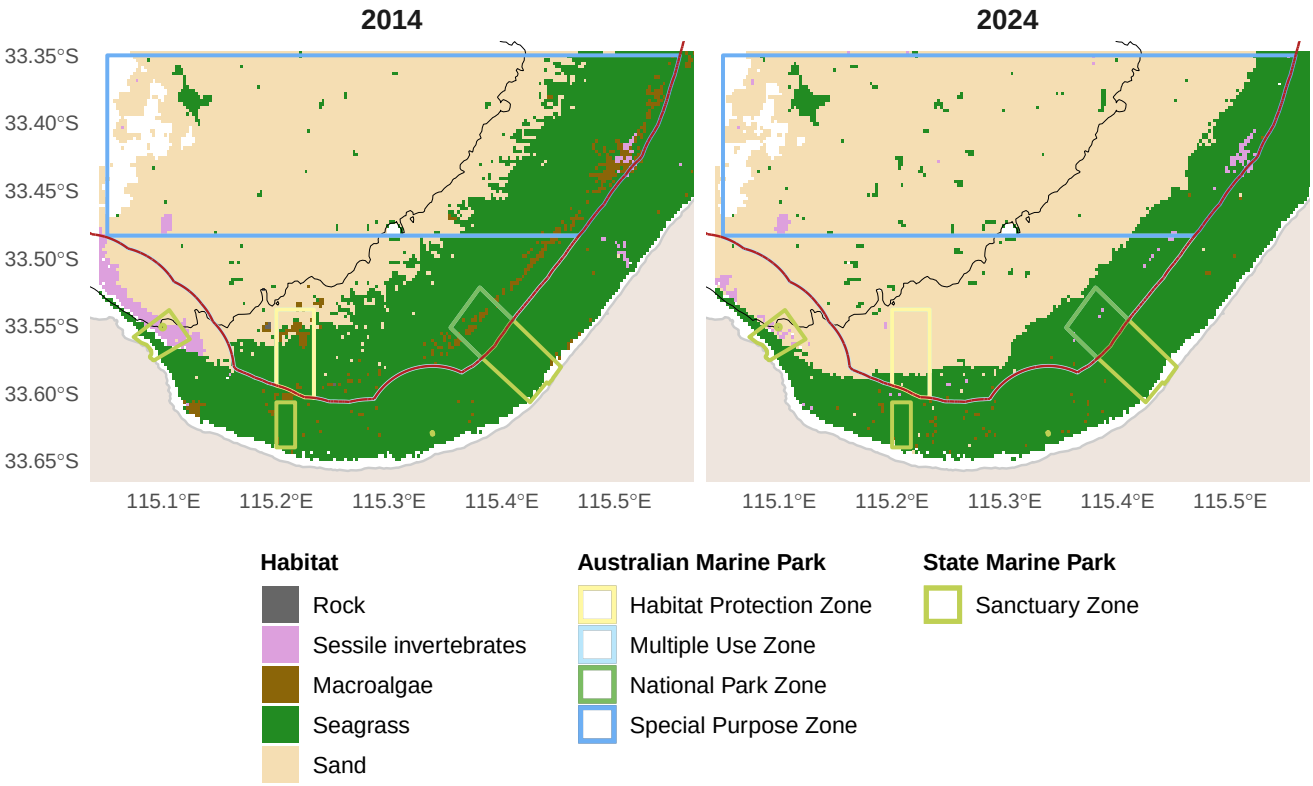


Figure A.2: Predicted dominant habitat type from stereo-BRUV groundtruthing.The green polygon represents the extent of no-take National Park Zones and State no-take sanctuary zones. State and Commonwealth marine park boundaries are shown. The red line delimits state and commonwealth waters. Bathymetric contour at 30 m is shown.

Habitat distribution and probability

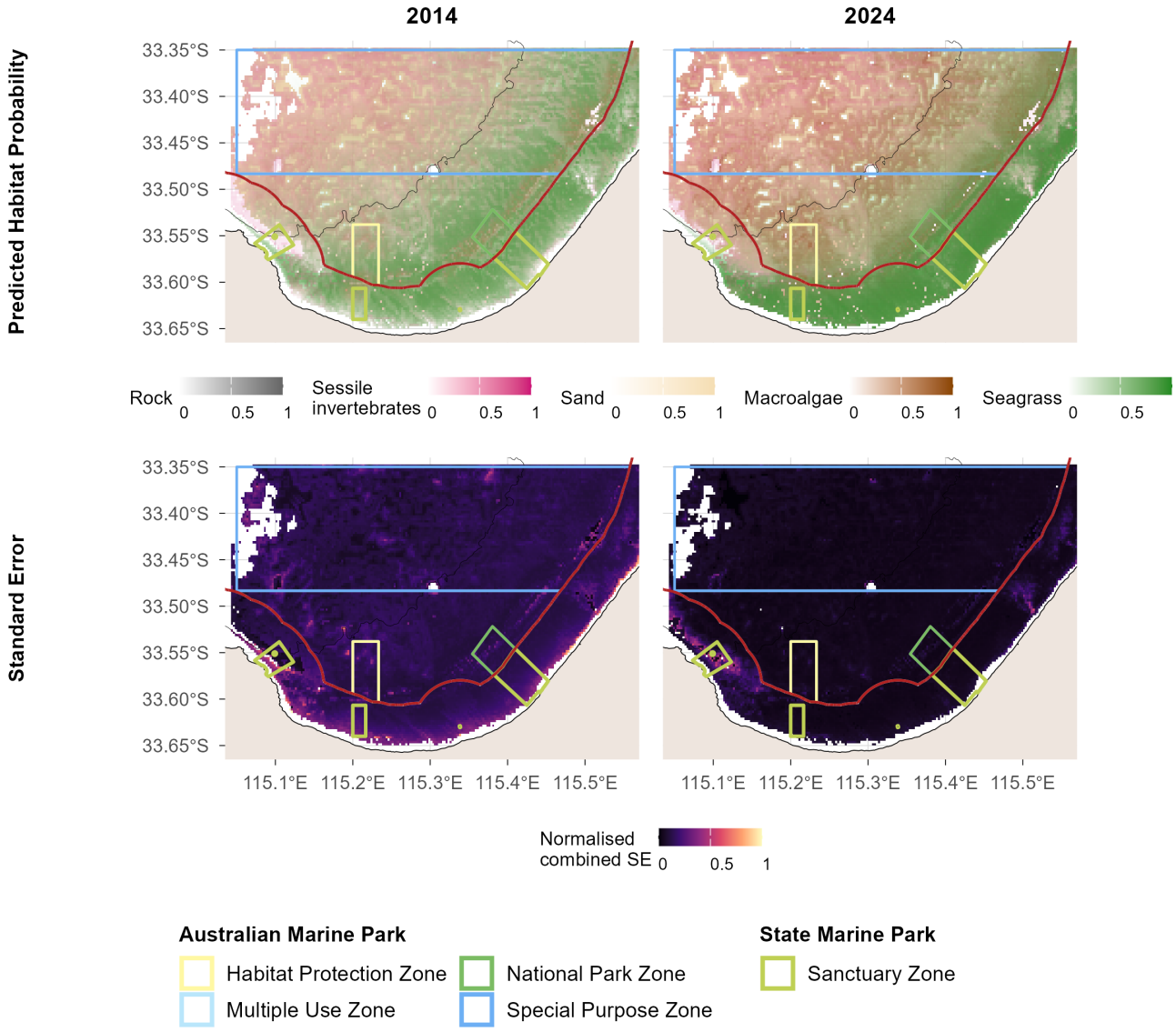


Figure A.3: Predicted habitat distribution probability and uncertainty from stereo-BRUV groundtruthing. The green polygon represents the extent of no-take National Park Zones. Commonwealth marine park boundaries are shown. The red line delimits state and commonwealth waters. Bathymetric contour at 30 m is shown.

Functional reef distribution and probability

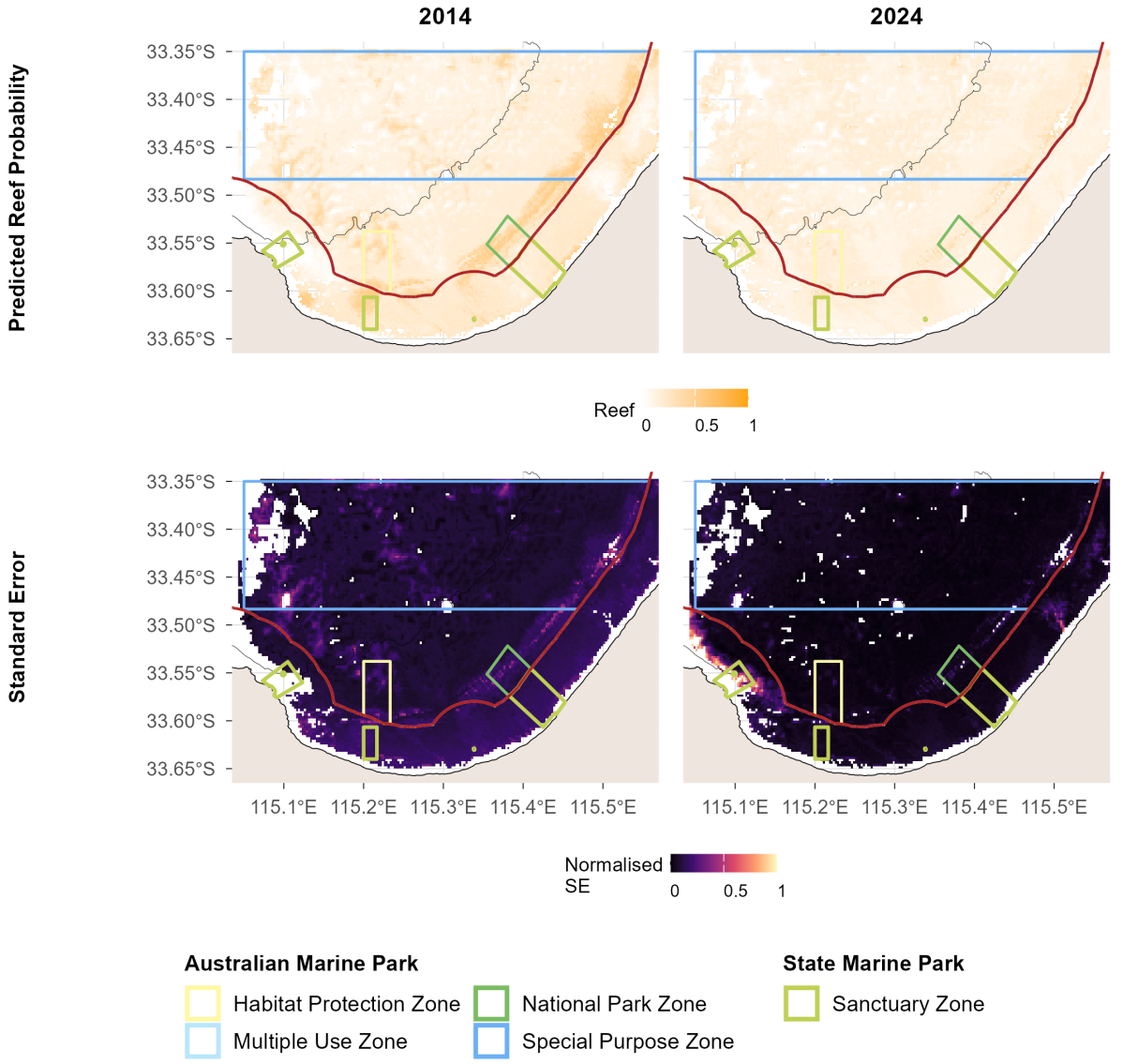


Figure A.4: Predicted reef distribution probability and uncertainty from stereo-BRUV groundtruthing.The green polygon represents the extent of no-take National Park Zones. Commonwealth marine park boundaries are shown. The red line delimits state and commonwealth waters. Bathymetric contour at 30 m is shown.

Benchmarks of benthic natural values

Seagrass

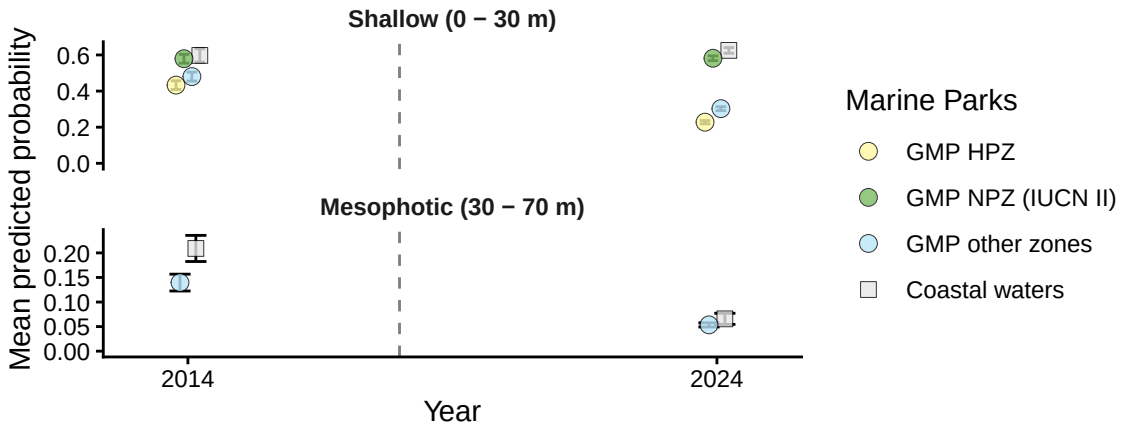


Figure A.5.1: Seagrass: temporal plots by depth contour from stereo-BRUV groundtruthing. Dashed vertical line indicates the date of establishment of the marine park. GMP = GMP (Commonwealth), HPZ = Habitat Protection Zone, NPZ = National Park Zone, NCMP = Ngari Capes Marine Park (State), SZ = Sanctuary Zone.

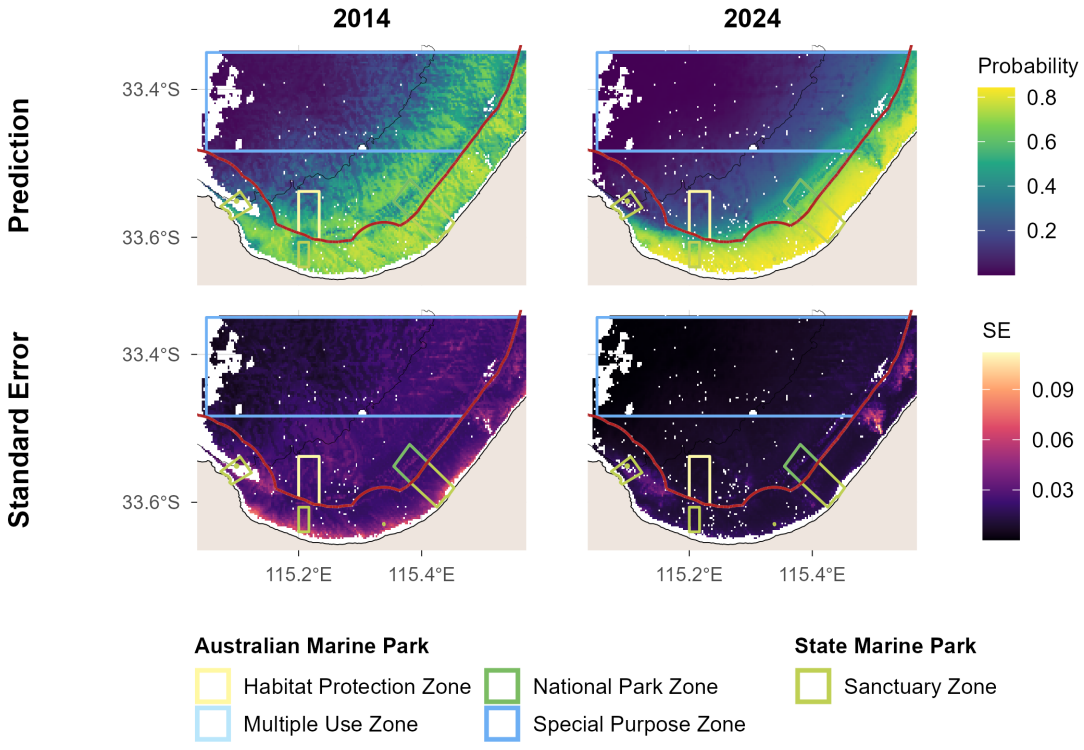


Figure A.5.2: Seagrass: spatial plots of distribution probability and uncertainty from stereo-BRUV groundtruthing. State and Commonwealth marine park boundaries are shown. The red line delimits state and commonwealth waters. Bathymetric contour at 30 m is shown.

Macroalgae

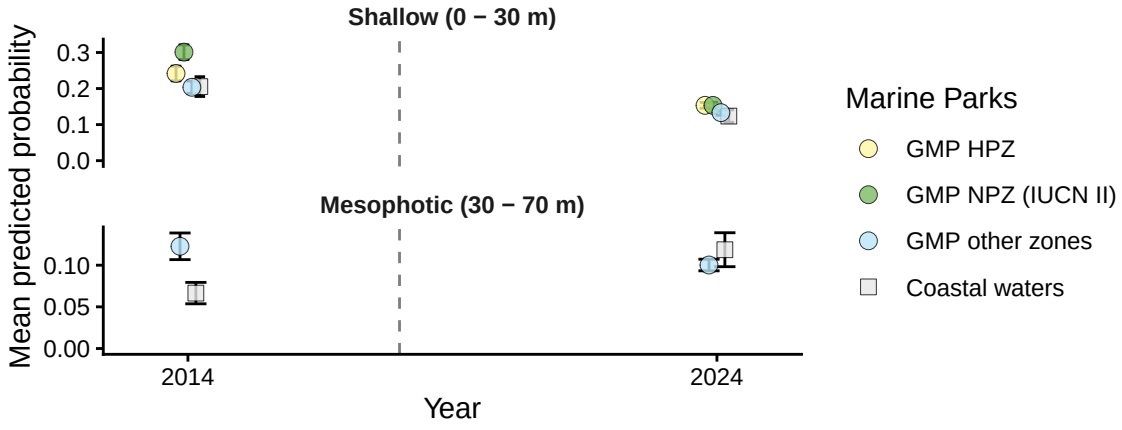


Figure A.6.1: Macroalgae: temporal plots by depth contour from stereo-BRUV groundtruthing. Dashed vertical line indicates the date of establishment of the marine park. GMP = GMP (Commonwealth), HPZ = Habitat Protection Zone, NPZ = National Park Zone, NCMP = Ngari Capes Marine Park (State), SZ = Sanctuary Zone.

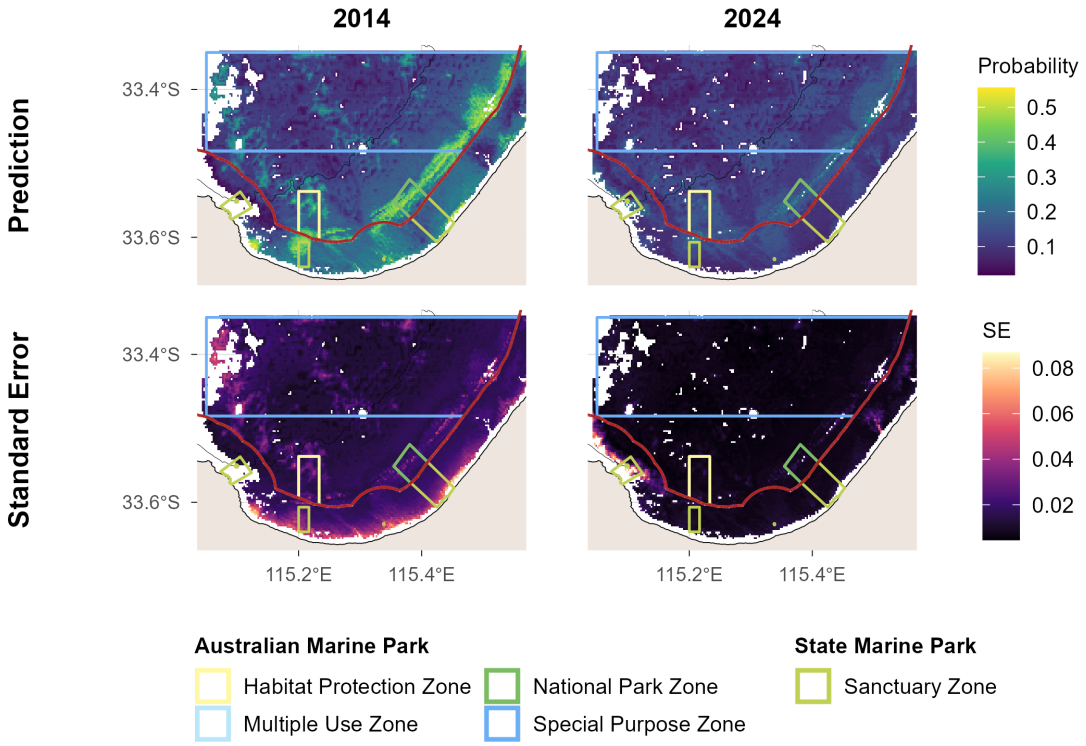


Figure A.6.2: Macroalgae: spatial plots of distribution probability and uncertainty from stereo-BRUV groundtruthing. State and Commonwealth marine park boundaries are shown. The red line delimits state and commonwealth waters. Bathymetric contour at 30 m is shown.

Sessile invertebrates

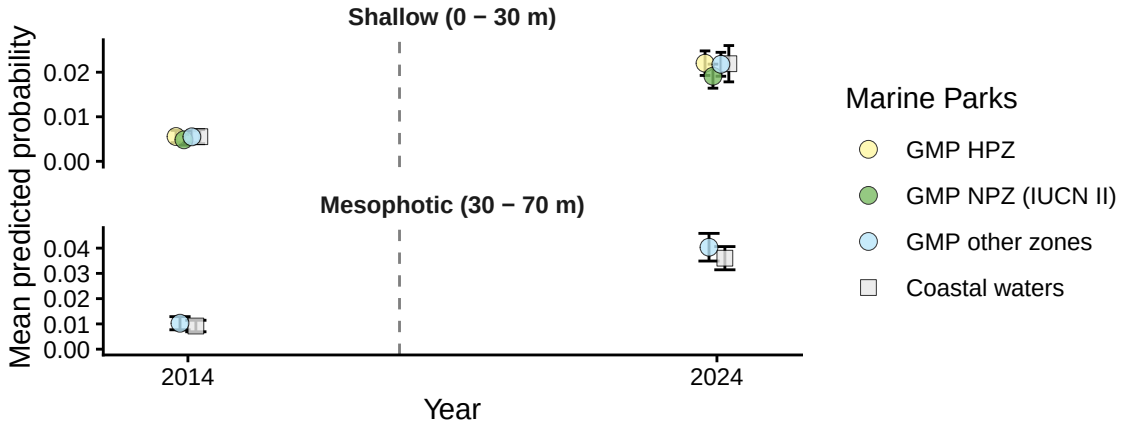


Figure A.7.1: Sessile invertebrates: temporal plots by depth contour from stereo-BRUV groundtruthing. Dashed vertical line indicates the date of establishment of the marine park. GMP = GMP (Commonwealth), HPZ = Habitat Protection Zone, NPZ = National Park Zone, NCMP = Ngari Capes Marine Park (State), SZ = Sanctuary Zone.

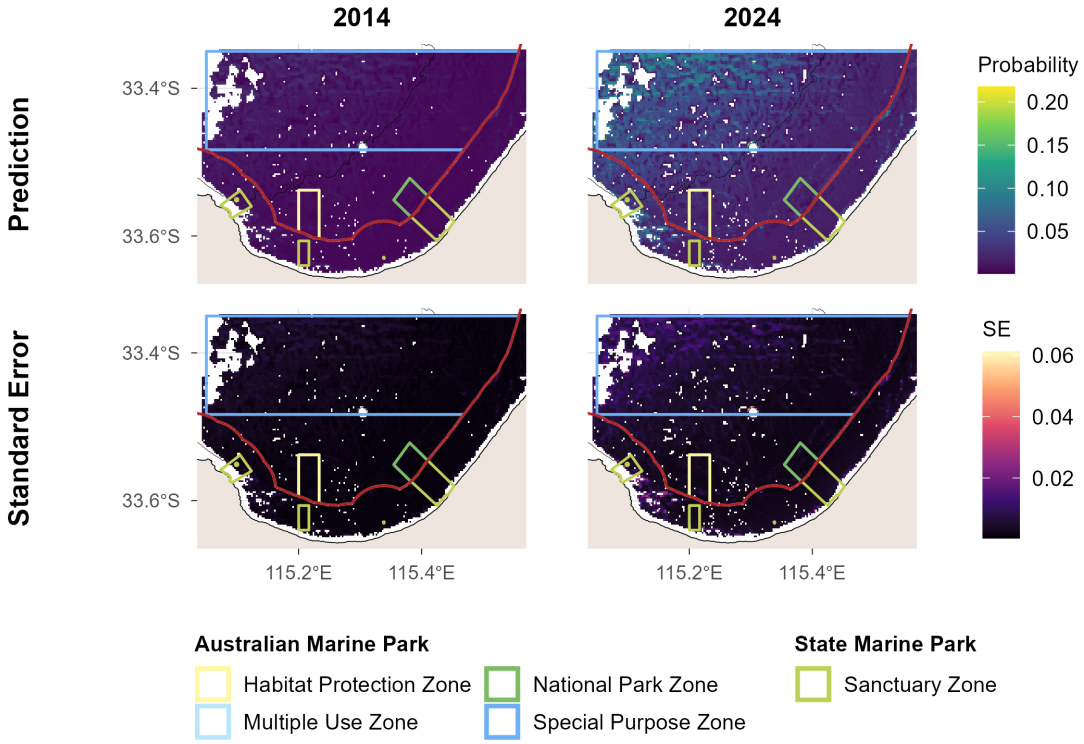


Figure A.7.2: Sessile invertebrates: spatial plots of distribution probability and uncertainty from stereo-BRUV groundtruthing. State and Commonwealth marine park boundaries are shown. The red line delimits state and commonwealth waters. Bathymetric contour at 30 m is shown.

Benchmarks of demersal fish natural values

Survey effort

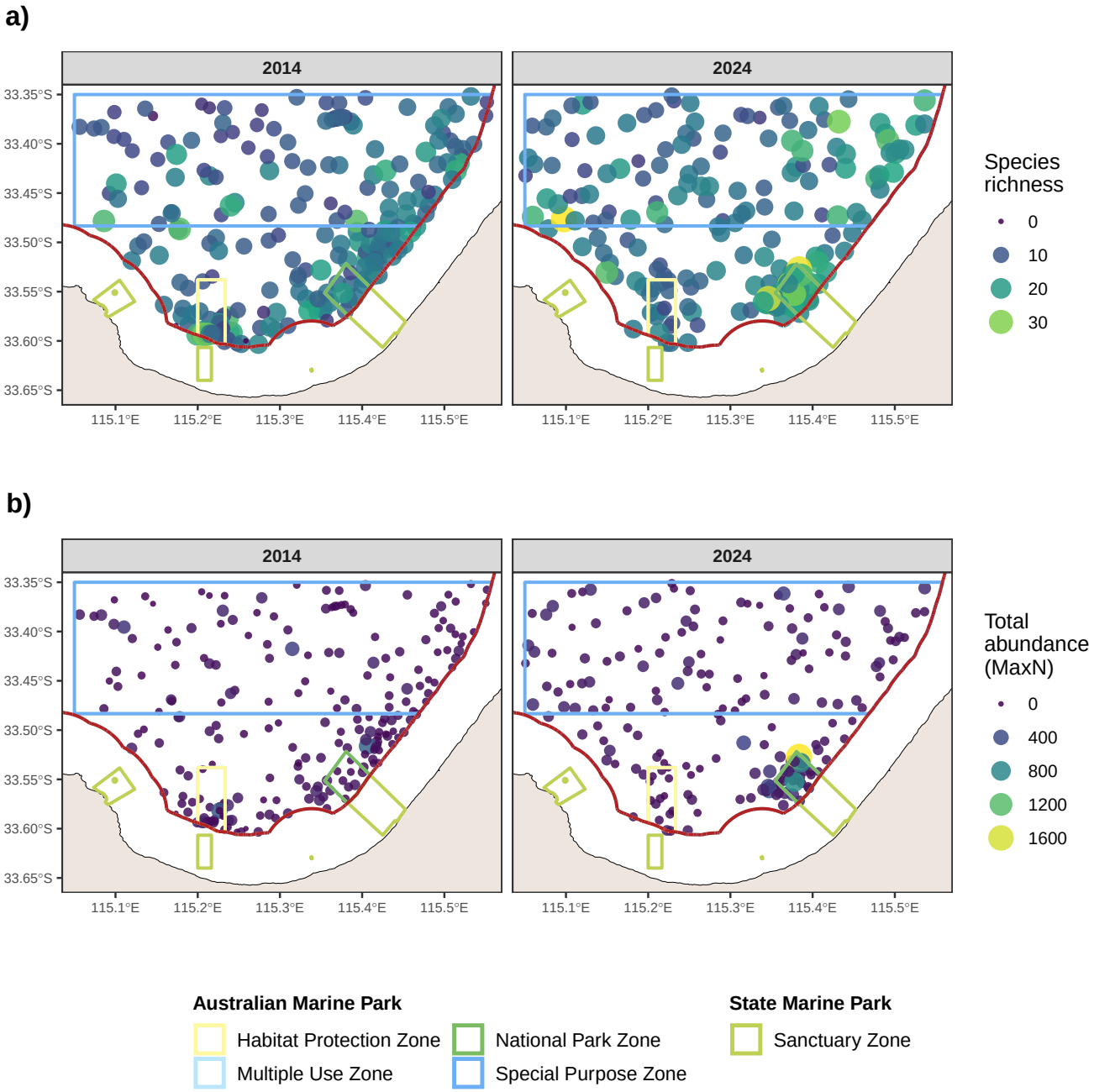


Figure A.8: Spatial distribution of observed species richness and total abundance (MaxN) per stereo-BRUV drop, pooled across all deployments. State and Commonwealth marine park boundaries are shown. The red line delimits state and commonwealth waters.

Species richness

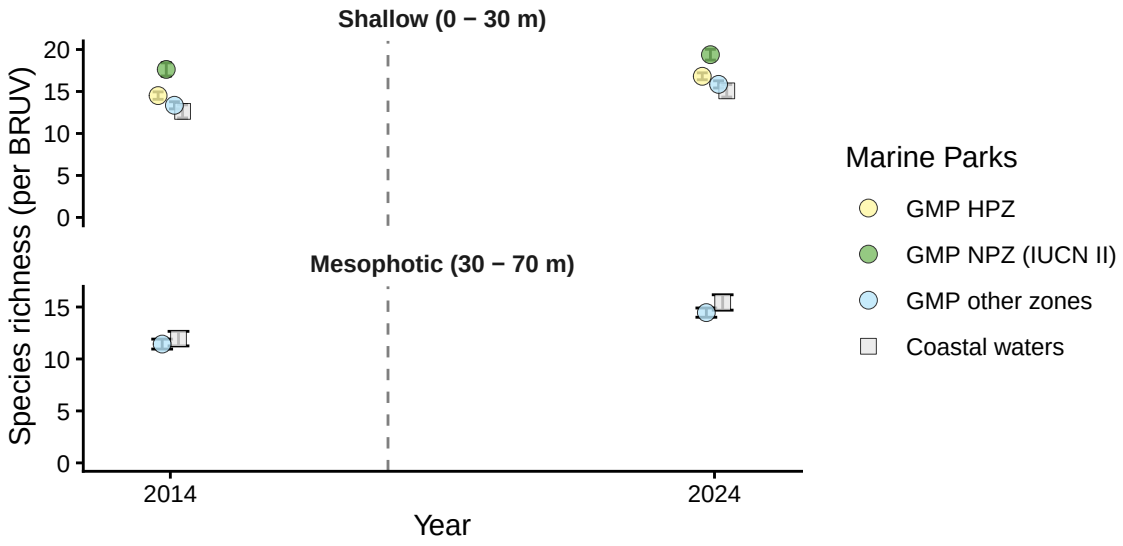


Figure A.9.1: Demersal fish - Species richness: benchmark plots by zone and depth contour from stereo-BRUV groundtruthing. Dashed vertical line indicates the date of establishment of the marine park. GMP = GMP (Commonwealth), HPZ = Habitat Protection Zone, NPZ = National Park Zone, NCMP = Ngari Capes Marine Park (State), SZ = Sanctuary Zone.

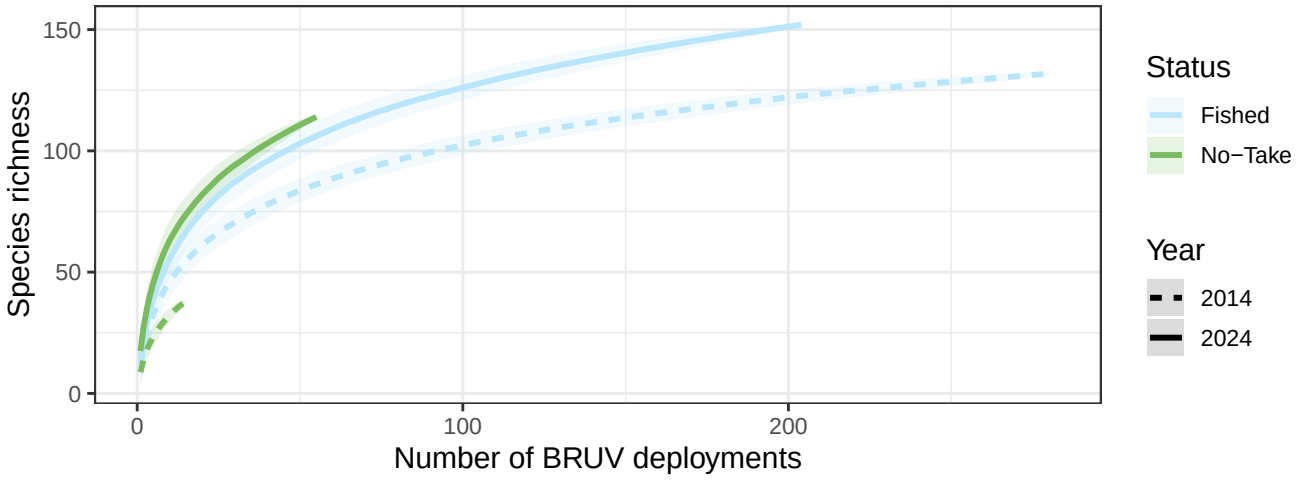


Figure A.9.2: Demersal fish - Species richness: species accumulation curve by zone type from stereo-BRUV groundtruthing.

Species richness

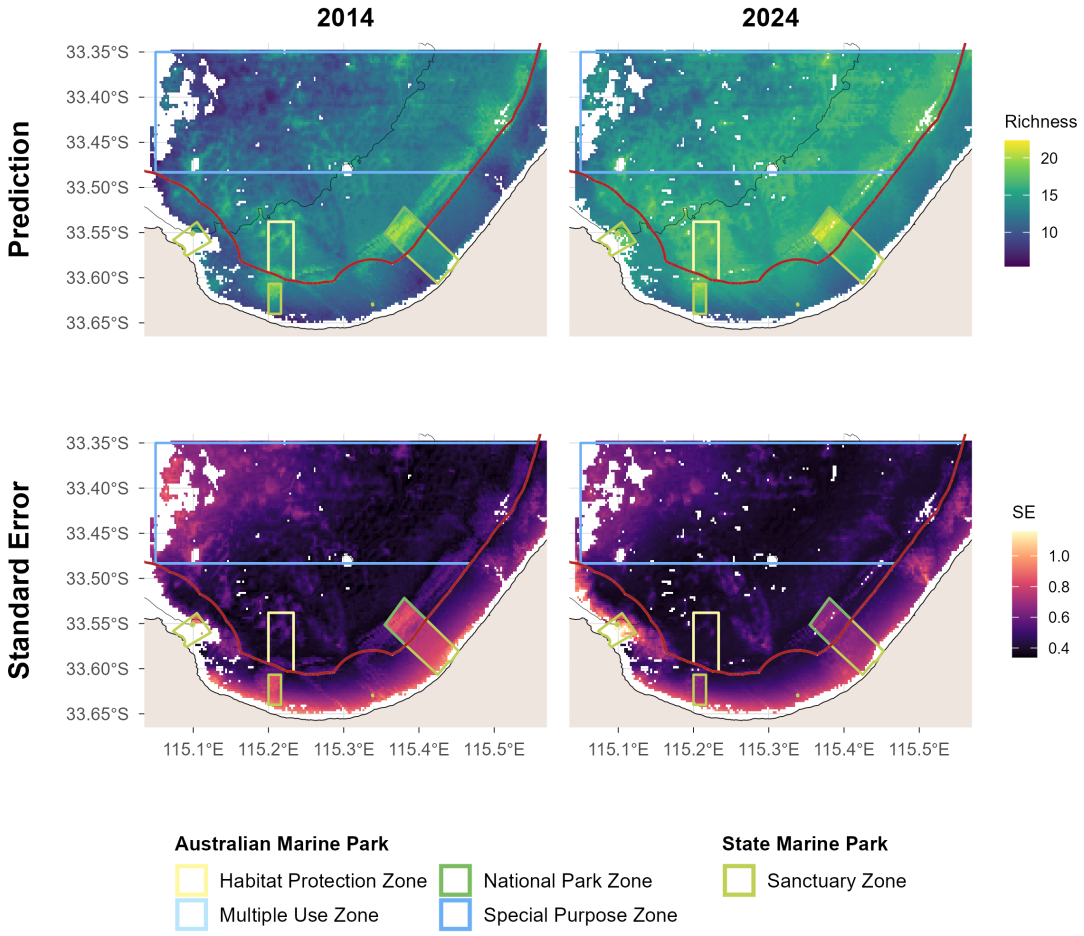


Figure A.9.3: Demersal fish - Species richness: spatial plots of distribution probability and uncertainty from stereo-BRUV groundtruthing. Commonwealth marine park boundaries are shown. The red line delimits state and commonwealth waters. Bathymetric contour at 30 m is shown.

Total abundance

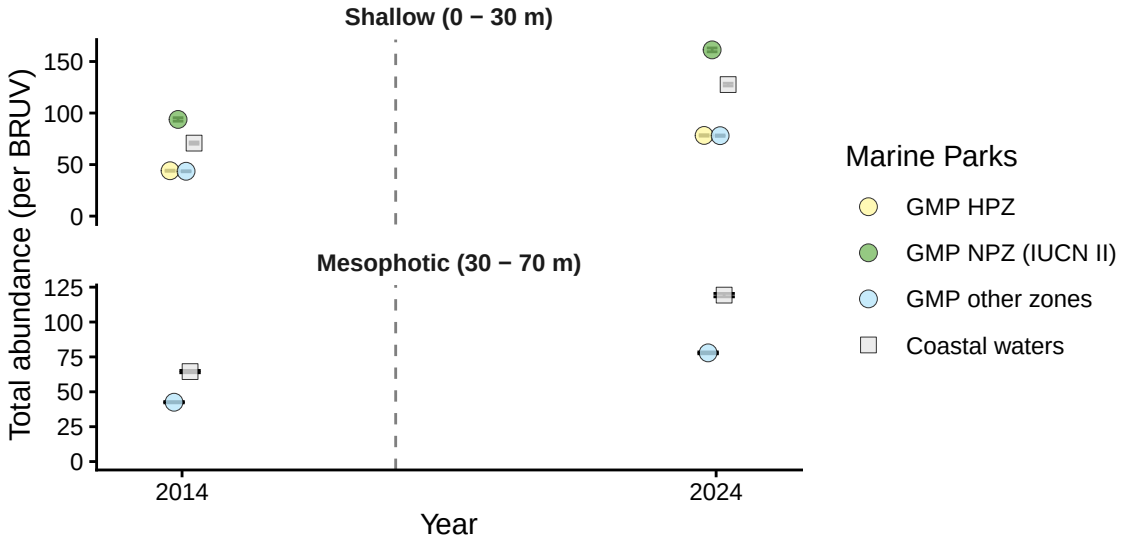


Figure A.10.1: Demersal fish - Total abundance: benchmark plots by zone and depth contour from stereo-BRUV groundtruthing. Dashed vertical line indicates the date of establishment of the marine park. GMP = GMP (Commonwealth), HPZ = Habitat Protection Zone, NPZ = National Park Zone, NCMP = Ngari Capes Marine Park (State), SZ = Sanctuary Zone.

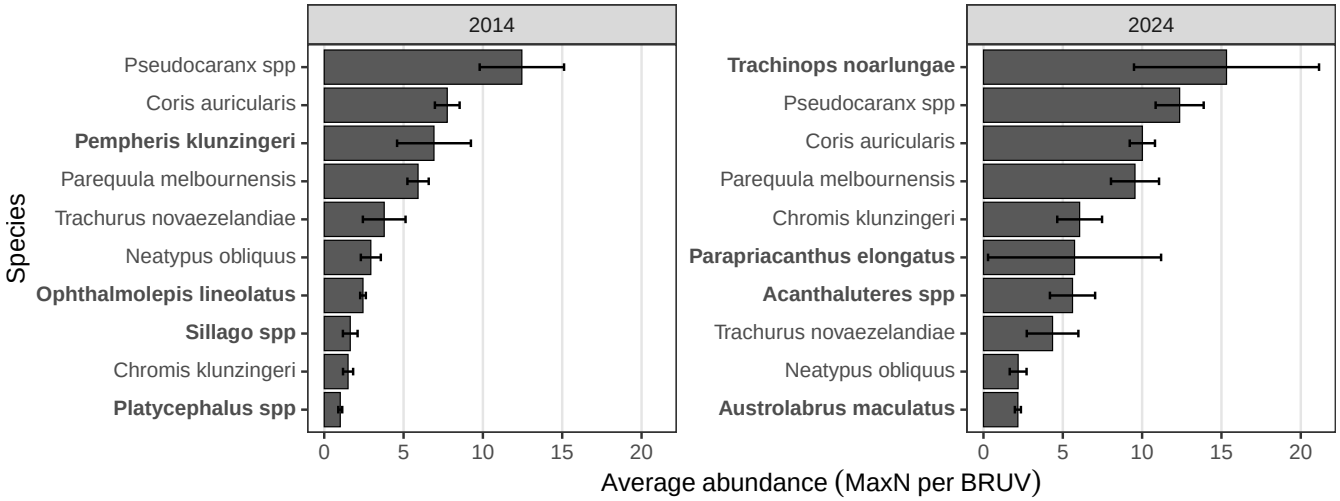


Figure A.10.2: Demersal fish - Total abundance: top 10 most abundant species (pooled across deployments) from stereo-BRUV groundtruthing. **Bold** species are unique within the 10 most abundant species by year.

Total abundance

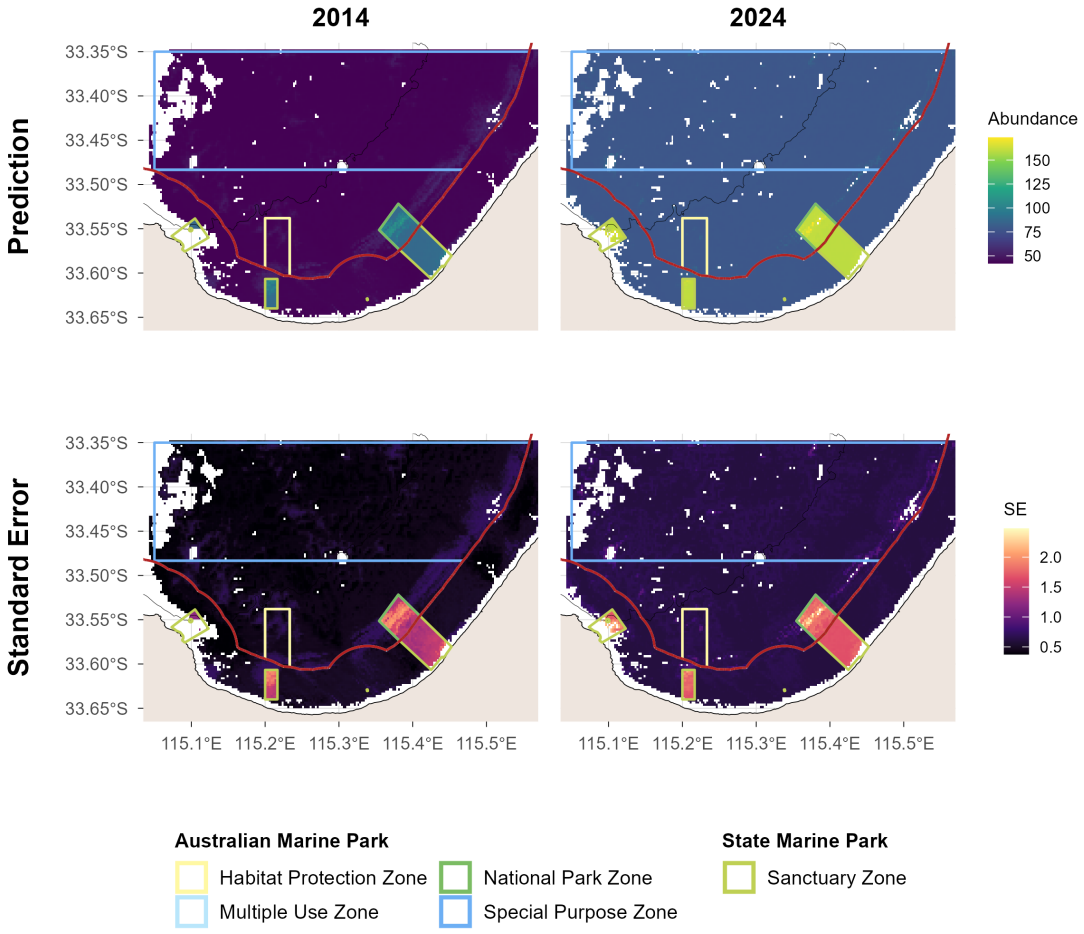


Figure A.10.3: Demersal fish - Total abundance: spatial plots of distribution probability and uncertainty from stereo-BRUV groundtruthing. Commonwealth marine park boundaries are shown. The red line delimits state and commonwealth waters. Bathymetric contour at 30 m is shown.

Large Reef Fish Index*

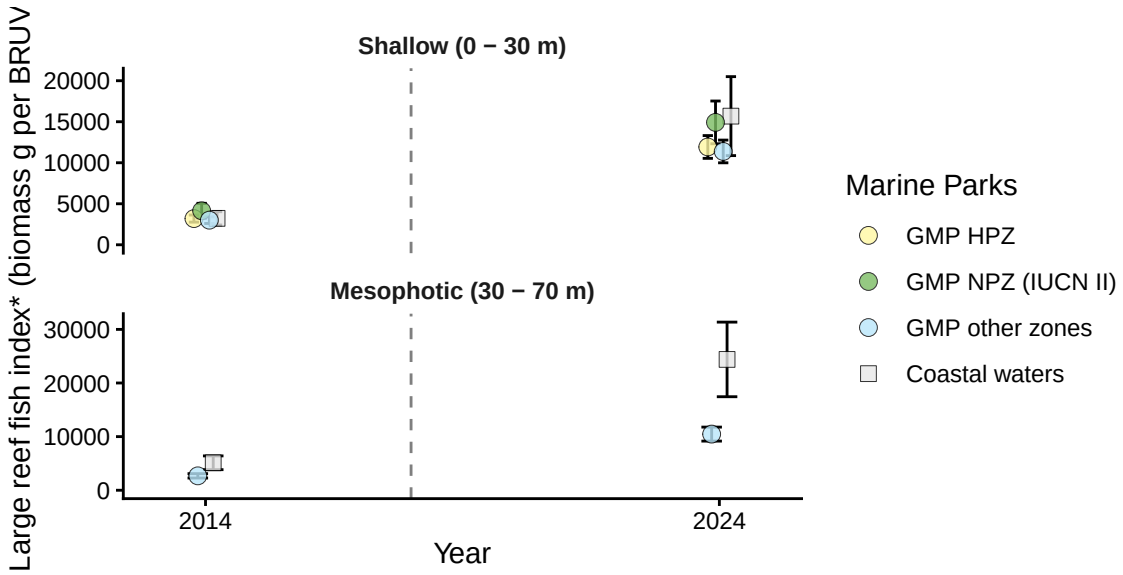


Figure A.11.1: Demersal fish - Large Reef Fish Index*: benchmark plots by zone and depth contour from stereo-BRUV groundtruthing. Dashed vertical line indicates the date of establishment of the marine park. GMP = GMP (Commonwealth), HPZ = Habitat Protection Zone, NPZ = National Park Zone, NCMP = Ngari Capes Marine Park (State), SZ = Sanctuary Zone. *Large Reef Fish Index representing the biomass of all fish greater than 20 cm (B20) has been adapted here for stereo-BRUV data by removing sharks and rays.

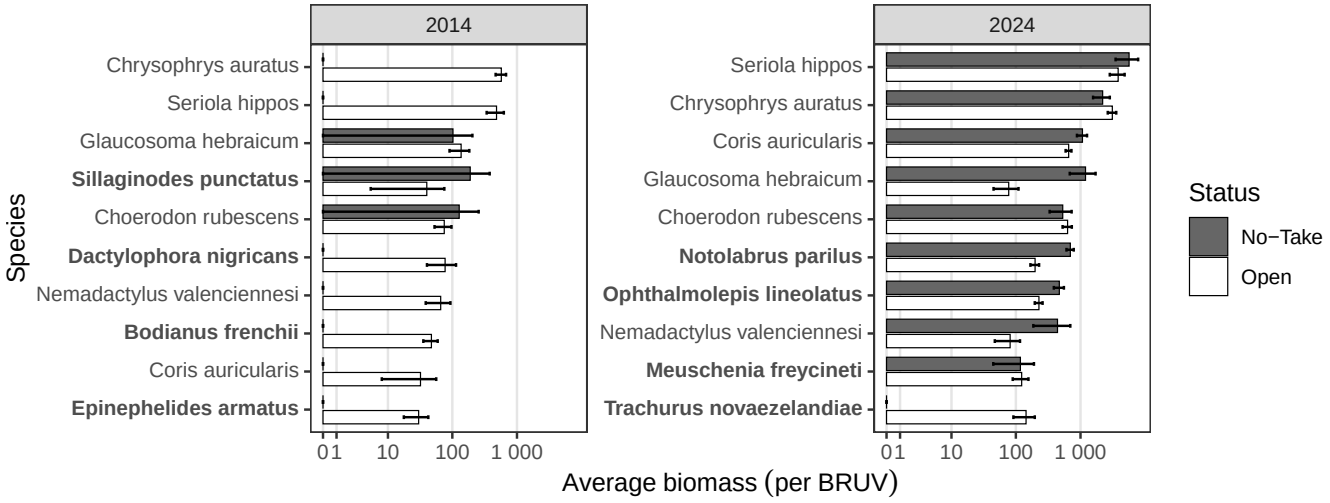


Figure A.11.2: Demersal fish - Large Reef Fish Index*: top 10 greatest biomass species by status (pooled across deployments) from stereo-BRUV groundtruthing. **Bold** species are unique within the 10 greatest biomass species by year.

Large Reef Fish Index*

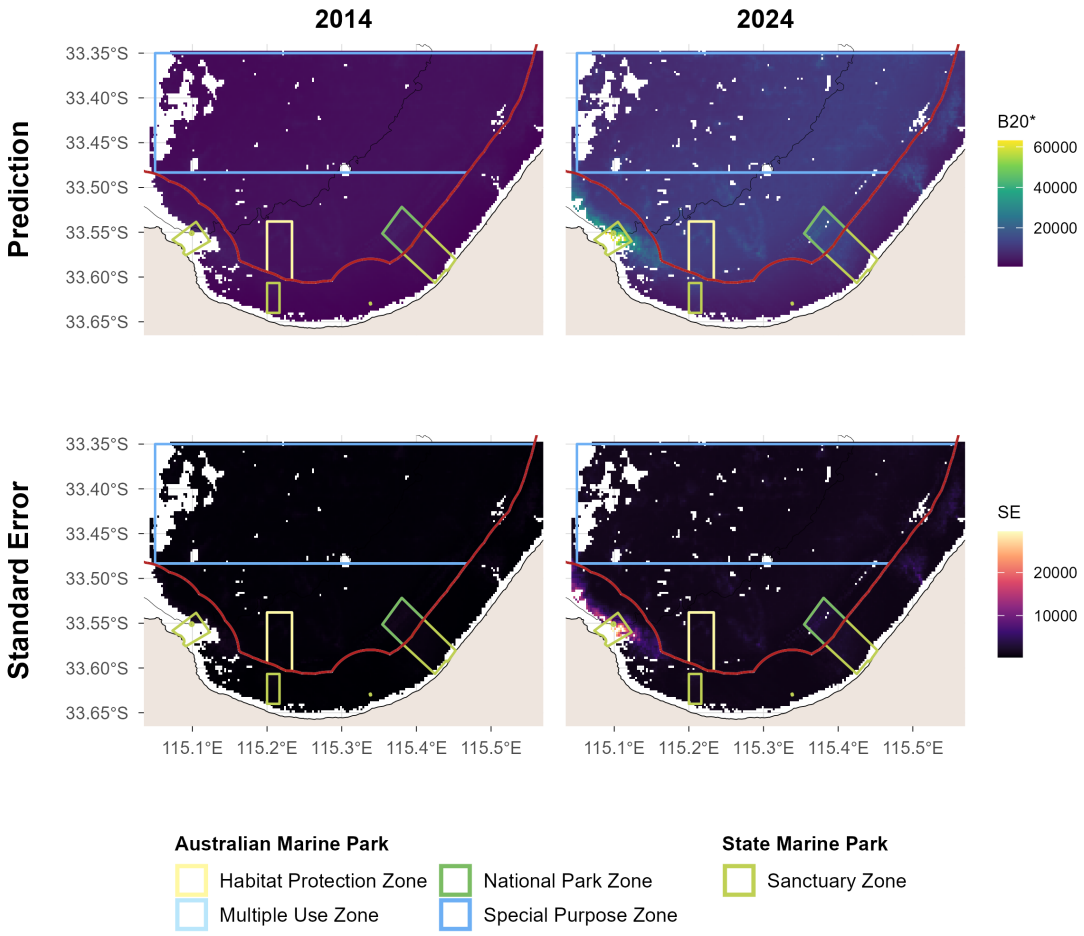


Figure A.11.3: Demersal fish - Large Reef Fish Index*: spatial plots of distribution probability and uncertainty from stereo-BRUV groundtruthing. Commonwealth marine park boundaries are shown. The red line delimits state and commonwealth waters. Bathymetric contour at 30 m is shown.

Reef fish thermal index

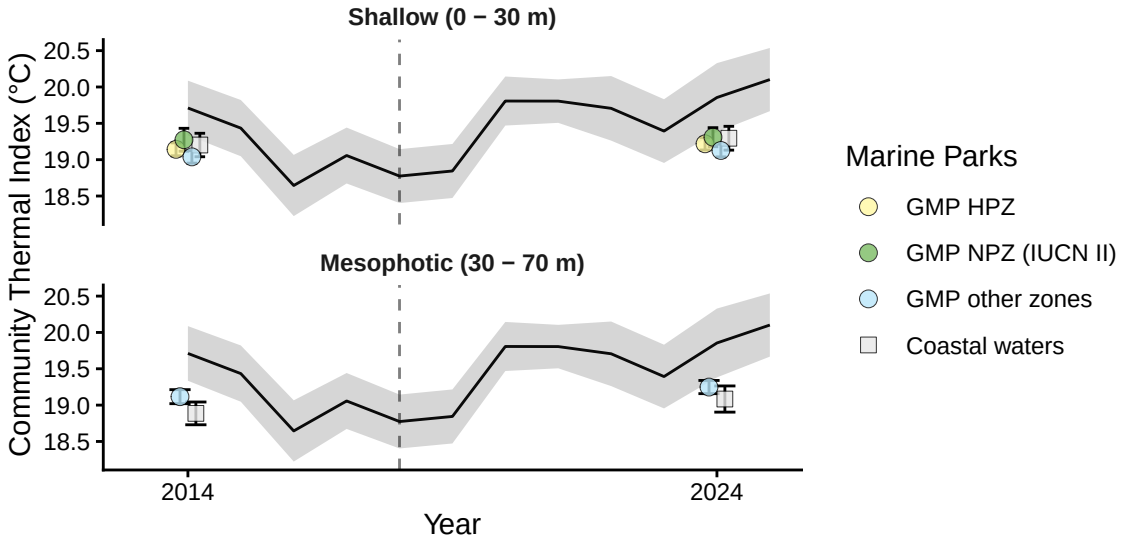


Figure A.12.1: Demersal fish - Reef Fish Thermal Index: benchmark plots by zone and depth contour, with monthly sea surface temperature (black line, grey band = SD) overlaid, from stereo-BRUV groundtruthing. Dashed vertical line indicates the date of establishment of the marine park. GMP = GMP (Commonwealth), HPZ = Habitat Protection Zone, NPZ = National Park Zone, NCMP = Ngari Capes Marine Park (State), SZ = Sanctuary Zone.

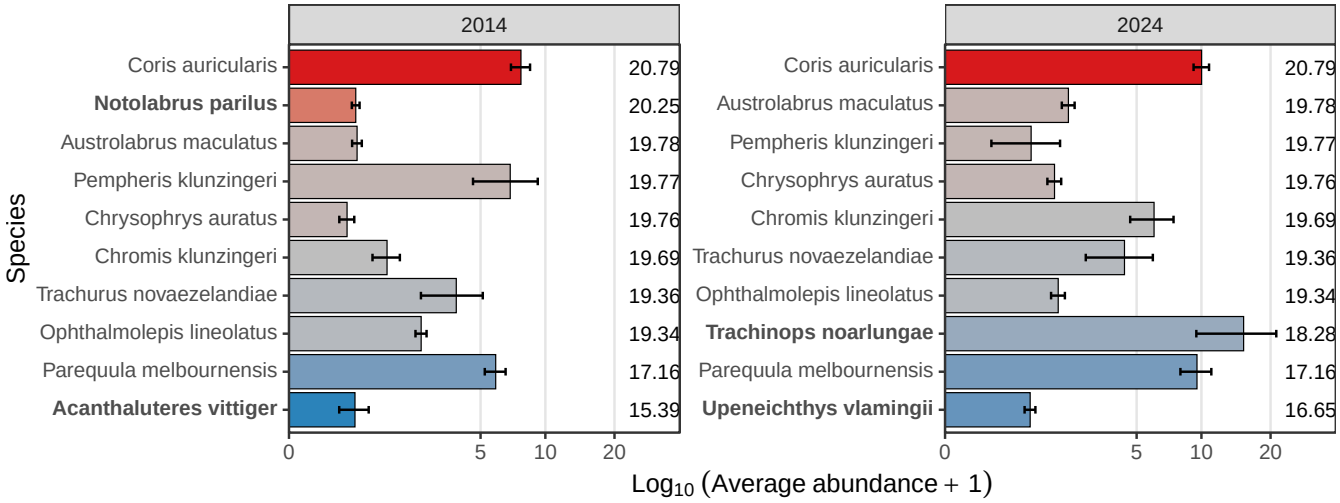


Figure A.12.2: Demersal fish - Reef Fish Thermal Index: species thermal niche of top 10 most abundant species (pooled across deployments) from stereo-BRUV groundtruthing. **Bold** species are unique within the 10 most abundant species by year.

Reef fish thermal index

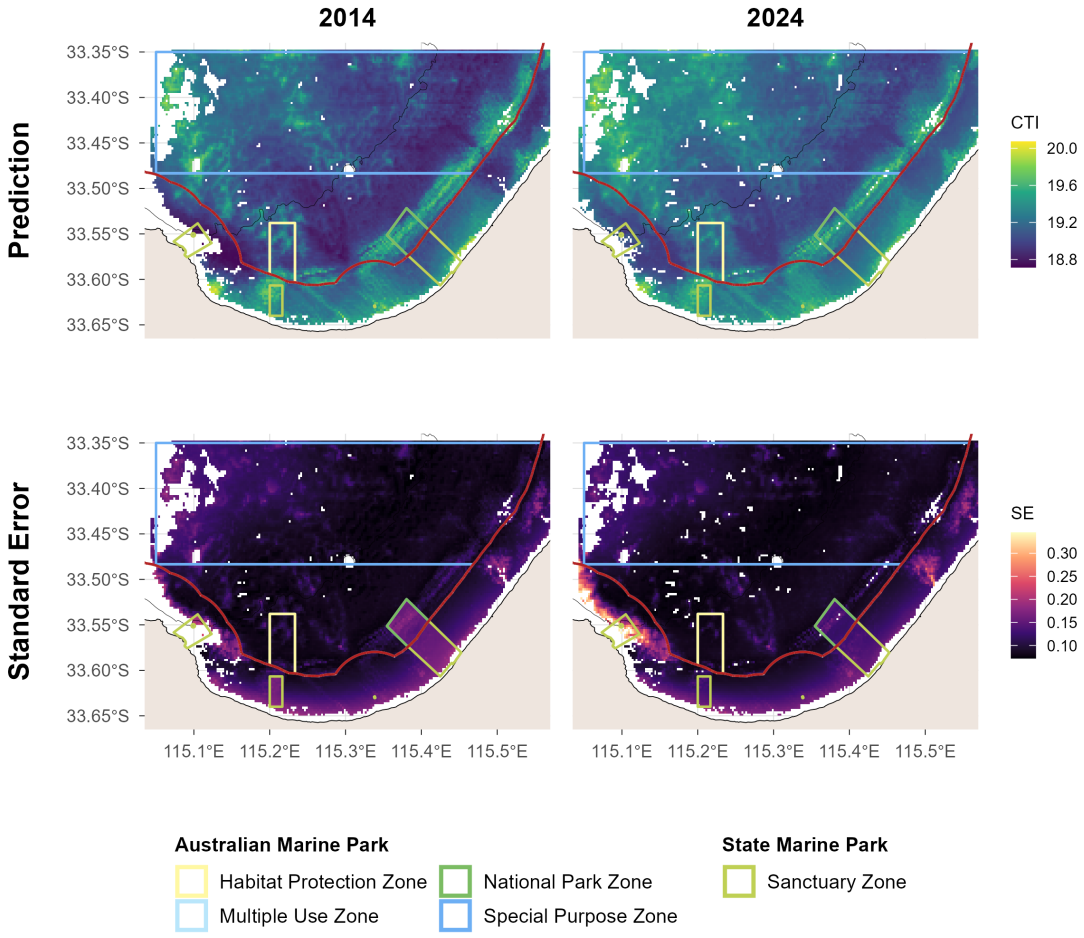


Figure A.12.3: Demersal fish - Reef Fish Thermal Index: spatial plots of distribution probability and uncertainty from stereo-BRUV groundtruthing. Commonwealth marine park boundaries are shown. The red line delimits state and commonwealth waters. Bathymetric contour at 30 m is shown.

Threatened species

Table A.2: Threatened species recorded in stereo-BRUV imagery across all survey years, showing IUCN Red List category, the number of samples each species was recorded in, and the total abundance (MaxN) summed across those samples. *This species is also listed as 'Conservation dependent' on the EPBC Act List of Threatened Fauna. ^This species is also listed as 'Vulnerable' on the EPBC Act List of Threatened Fauna.

Species	Common name	Family	IUCN category	Samples	Total MaxN	Years recorded
Bodianus frenchii	Foxfish	Labridae	Near Threatened	50	59	2014, 2024
Epinephelides armatus	Breaksea Cod	Serranidae	Near Threatened	26	30	2014, 2024
Carcharhinus brevipinna	Spinner Shark	Carcharhinidae	Vulnerable	11	23	2014, 2024
Carcharhinus obscurus	Dusky Whaler	Carcharhinidae	Endangered	10	10	2024
Galeocerdo cuvier	Tiger Shark	Carcharhinidae	Near Threatened	6	6	2014, 2024
Sepia apama	Giant cuttlefish	Sepiidae	Near Threatened	6	6	2024
*Sphyrna zygaena**	Smooth Hammerhead	Sphyrnidae	Vulnerable	6	6	2014, 2024
Achoerodus gouldii	Western Blue Groper	Labridae	Vulnerable	5	5	2014, 2024
*Thunnus maccoyii**	Southern Bluefin Tuna	Scombridae	Endangered	2	2	2024
Bathytoshia lata	Black Stingray	Dasyatidae	Vulnerable	1	1	2024
Carcharhinus brachyurus	Bronze Whaler	Carcharhinidae	Vulnerable	1	1	2024
Carcharhinus limbatus	Common Blacktip Shark	Carcharhinidae	Vulnerable	1	2	2024
Isurus oxyrinchus	Shortfin Mako	Lamnidae	Endangered	1	1	2014
Rhynchobatus australiae	Whitespotted Guitarfish	Rhinidae	Critically Endangered	1	1	2024
*Sphyrna lewini**	Scalloped Hammerhead	Sphyrnidae	Critically Endangered	1	1	2024

Threatened shark and ray species

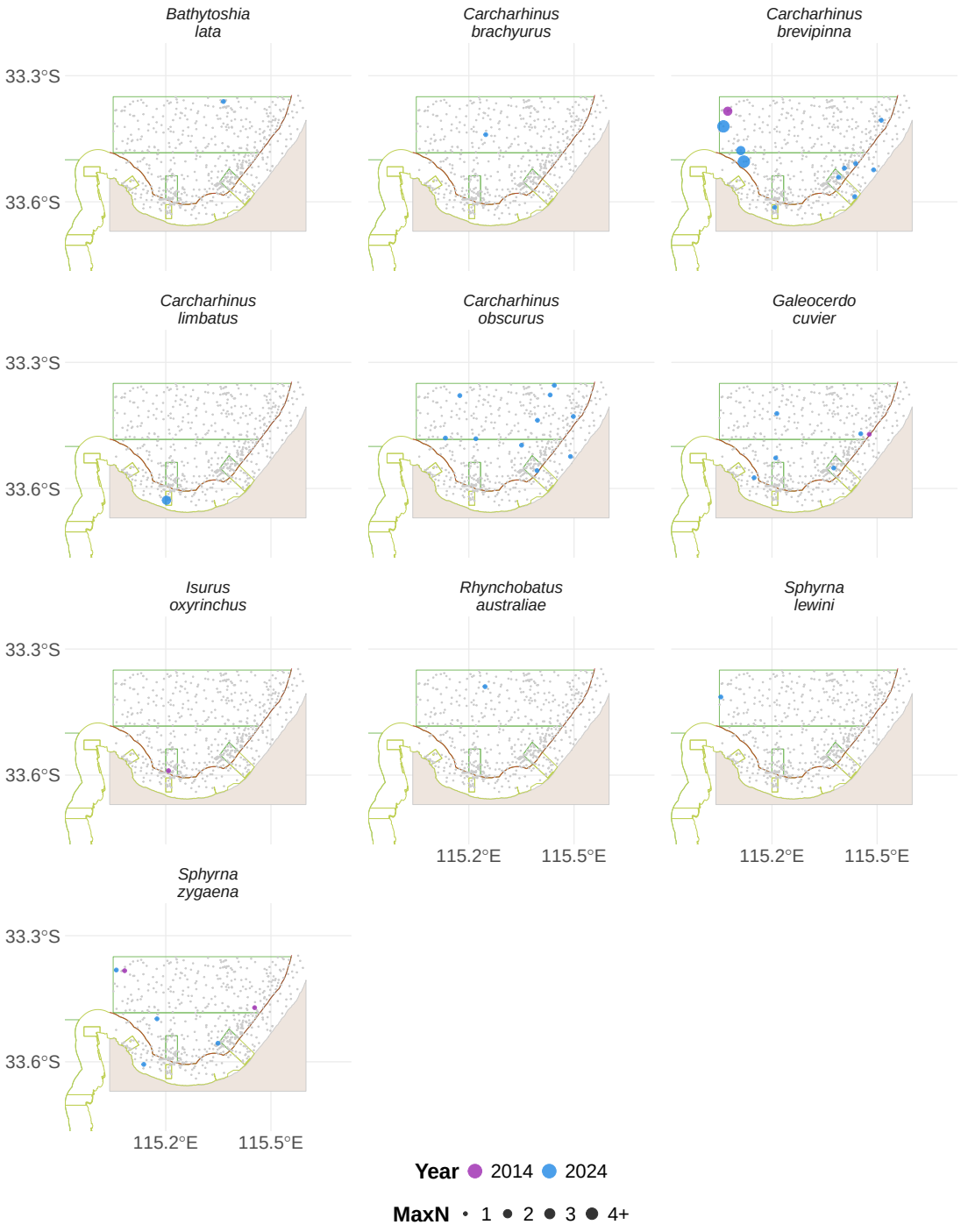


Figure A.13.1: Occurrence of threatened, IUCN Red List category, sharks and rays in stereo-BRUV imagery. Points are scaled by abundance (MaxN) and coloured by survey year; grey points show every stereo-BRUV deployment. State and Commonwealth marine park boundaries are shown. The red line delimits state and commonwealth waters.

Other shark and ray species

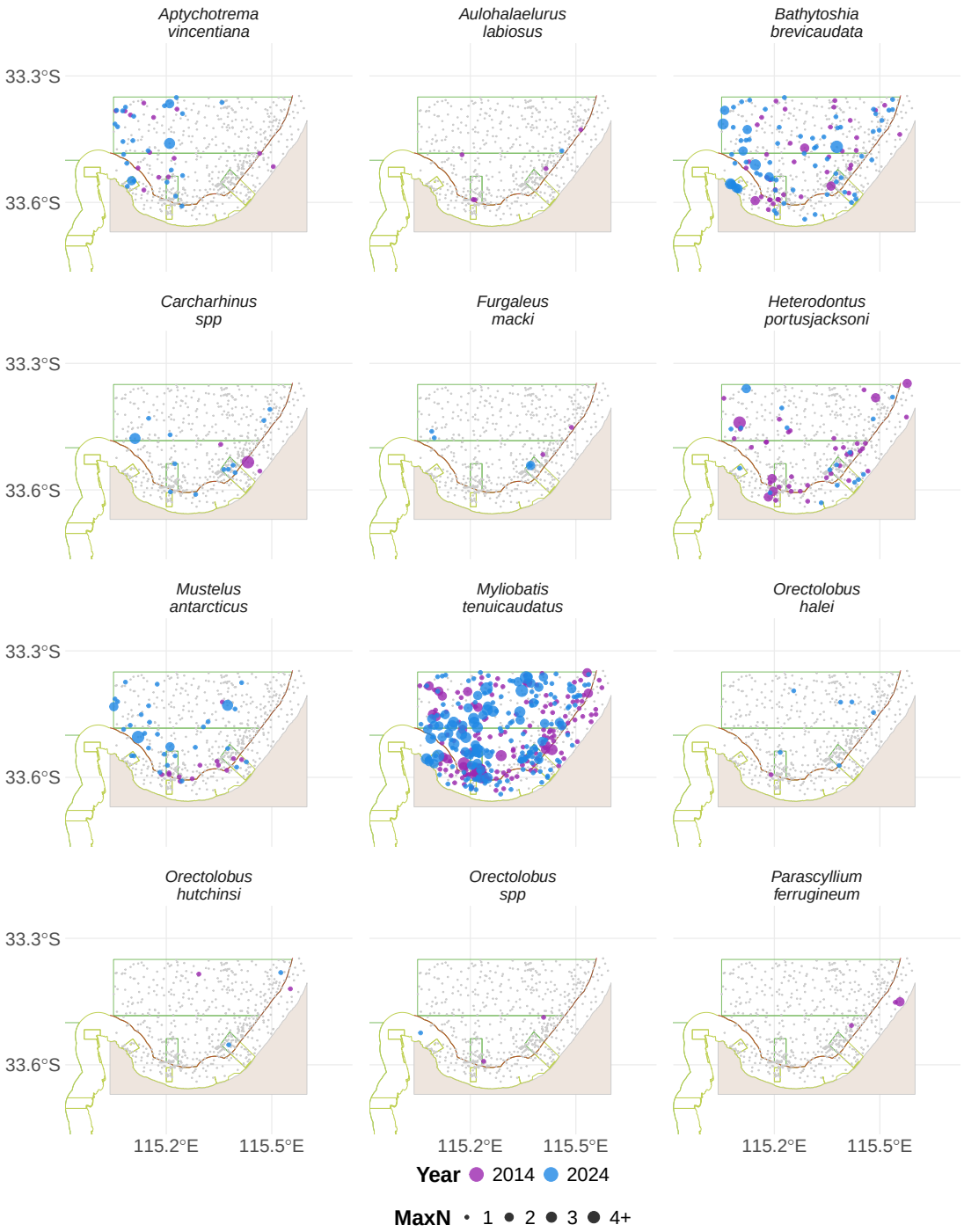


Figure A.13.2: Occurrence of shark and ray species that are not threatened in stereo-BRUV imagery. Points are scaled by abundance (MaxN) and coloured by survey year; grey points show every stereo-BRUV deployment, so a panel with no coloured points is a genuine absence rather than an absence of sampling. State and Commonwealth marine park boundaries are shown. The red line delimits state and commonwealth waters.

Other shark and ray species

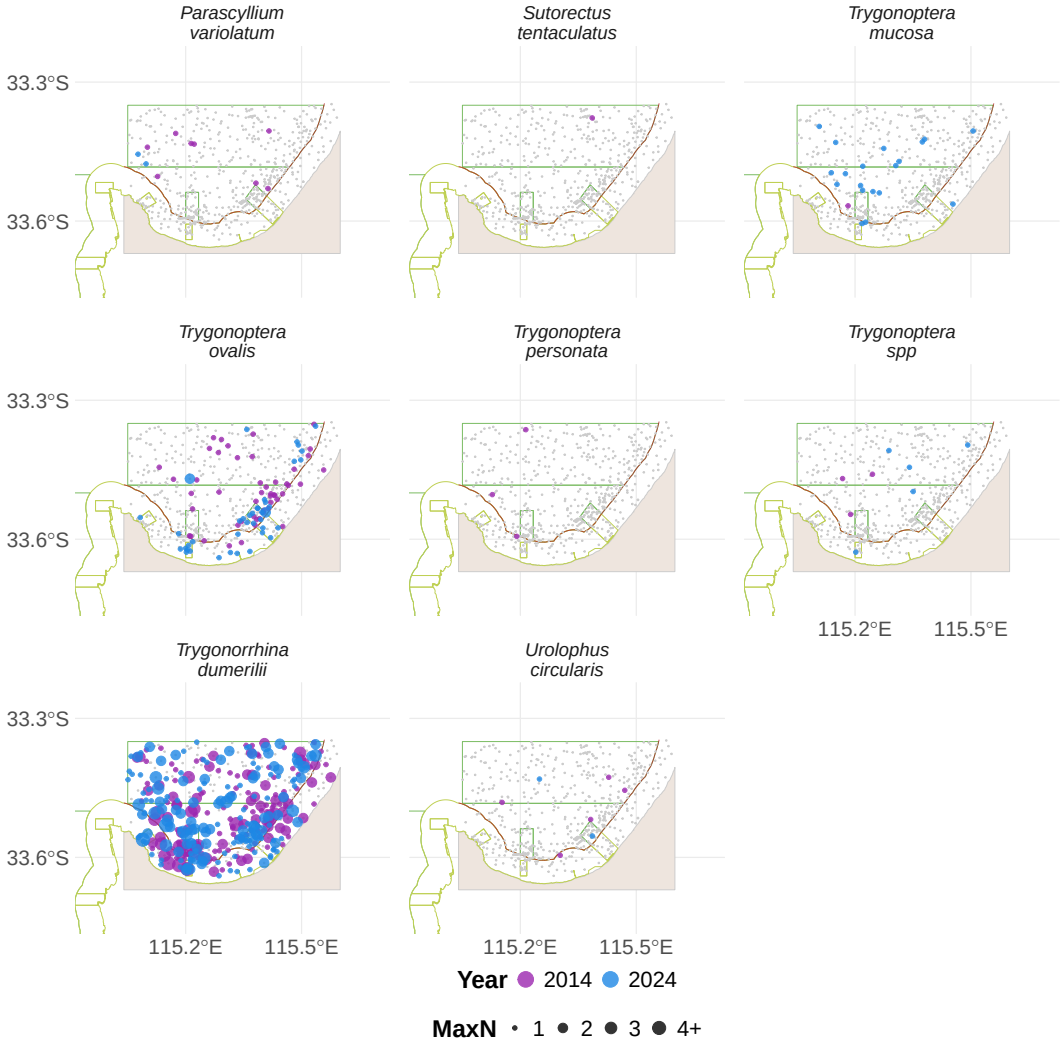


Figure A.13.3: Occurrence of shark and ray species that are not threatened in stereo-BRUV imagery (continued). Points are scaled by abundance (MaxN) and coloured by survey year; grey points show every stereo-BRUV deployment, so a panel with no coloured points is a genuine absence rather than an absence of sampling. State and Commonwealth marine park boundaries are shown. The red line delimits state and commonwealth waters.